

N K Vamsi Krishna

✉ nkvk@seas.upenn.edu ☎ +1 267-408-4672 🌐 vamsiverse.com 📺 trishulam in vamsikrishnank

EDUCATION

University of Pennsylvania

Philadelphia, PA, USA

Master of Science in Engineering (MSE), Data Science; GPA: 3.91/4.0

August 2025 – May 2027

Courses: CIS 5210 Artificial Intelligence, CIS 7000 Agentic AI, CIS 5190 Applied Machine Learning, CIS 5450 Big Data Analytics

Highlights: Teaching Assistant for CIS 5150 & CIS 5110 — Linear Algebra & Optimization; Theory of Computation

Indian Institute of Technology, Madras

Chennai, India

Bachelor of Science – Data Science & Applications; GPA: 3.91/4.0

Jan 2021 – Apr 2025

Highlights: Top 1% GPA; Academic Merit Certificate. Coursework included Deep Learning, Reinforcement Learning.

National University of Singapore

Singapore

Global Academic Internship Program — Data Analytics Using Deep Learning

June 2023

On-campus 3-week NUS SoC program; seminars + capstone project; **Best performer in a 100-member cohort.**

PROFESSIONAL EXPERIENCE

Amazon AGI

Sunnyvale, CA, USA

Incoming Software Development Engineer Intern, ML Infrastructure & Tooling

Sep 2026 – Dec 2026

Google - YouTube Trust & Safety, 1P GenAI

Mountain View, CA, USA

Software Engineering Intern

May 2026 – Aug 2026

- Benchmarked **9 architectures** and **8 weight-merging methods** (SLERP, LERP, TIES, MagMax) in **JAX** across Gemma-3 encoder classifiers and causal-decoder raters, establishing that consolidation must happen in **data space for encoders** and **weight space for decoders**, from gradient conflict and later-layer representation drift respectively.
- Handed off a **joint unmasked multi-task encoder** (shared violative head) at **98.33% macro ROC-AUC** and a **+8.5 pp recall gain** over per-policy experts at 95% precision, replacing a **4x GlassBox ensemble** at **1x cost** while lifting worst-policy recall from **6.49% to 62.86%**; unmasked labels contributed **+8.58 pp** average recall over masked training.
- Designed the team's **OOD adaptation playbook**: sequential **zero-replay fine-tuning** plus **SLERP 60/40 weight merging**, eliminating the 15% replay buffer, cutting adaptation compute **over 60%**, and holding ID AUC at **91.3%**, 3 pts above full retraining with no legacy-policy regression; **first-author paper under review, NeurIPS 2026 workshop**.
- Built a long-horizon human-in-the-loop **AutoML agent** with persistent state that runs the model-update playbook end to end, curating OOD data, launching training, merging weights, and validating checkpoints against quality thresholds before promoting a production candidate, with **playbooks as extensible modules** so other engineers can onboard their workflows.

S2T AI - AI Powered Investigations

Singapore

Software Engineer

Dec 2024 – Aug 2025

- Owned the company's **agentic AI** from prototype to customers: a **Knowledge Graph-RAG** microservice searching documents and databases, plus an **Agentic AI Assistant** with multi-agent orchestration, tool-calling, and Generative UI.
- Developed an **AI ETL service** for natural-language queries on unstructured data and deployed **8 K8s microservices** through Azure DevOps; improved audio spoofing detection by **15%**.

Vessel Match (Startup Incubated at Nirmaan IIT Madras) 📺 Product Demo

Chennai, India

Founder & CEO

May 2024 – Jun 2025

- Founded and led a **full-stack AI platform** automating vessel/cargo matching and pre-fixture workflows; built an **event-driven microservices architecture** using AWS (Lambda, SQS, EC2), FastAPI, Next.js, Supabase, RAG, and MCP.
- Processed **200+ vessels** and **150+ cargo listings** daily in trials, forged **5 industry partnerships**, and secured INR 700,000 in **seed funding** plus \$2K AWS credits while leading all technical development, architecture, and AI efforts.

PROJECTS

ThinkSpace [📺 Demo 📺 Code 📺 Devpost]

Mar 2026

Built a **real-time proactive multimodal AI tutor** that teaches over voice and a shared editable canvas. Designed an autonomous canvas interpreter that watches learner activity and decides **when** to intervene, turning a passive voice assistant into a tutor that catches confusion as it happens; sessions resume across visits and export as recordings and lecture notes.

React · TypeScript · Python · FastAPI · Gemini Live · Google ADK · tldraw · MediaPipe · GCP

SGC SWE Agent [📺 Code 📺 Report]

Apr 2026

Built **State-Governed Checkpointing**, supervisor middleware that maintains a **Semantic Working State** of goals, hypotheses, and failures outside the LLM context and steers the agent on drift; lifted **SWE-bench Verified (mini)** resolution from **66.7% to 72.3%** (+11.2 pp Django) while cutting agent steps **23%**.

Python · LangGraph · Deep Agents · Docker · SWE-bench · GPT-5.5 · Gemini Flash 3

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Neo4j)

AI/ML: JAX, PyTorch, Model Merging, Continual Learning, Multi-task SFT, LoRA, RAG, Knowledge Graphs, MCP

Tools & Infra: FastAPI, React, Next.js, Docker, Kubernetes, Git, Azure DevOps, AWS, GCP, Supabase, Redis

ACHIEVEMENTS & LEADERSHIP

- Winner - 2026 Wharton Hack-AI-thon (WayWise)** [📺]: First among 61 teams; a system to generate review questions.
- Winner - HackPrinceton 2025 (MLH)** [📺]: Best (Overall, Practical AI Innovation, Use of Gemini API) among 195 teams
- Best Use of Gemini API - YHack 2026 (Ayana)** [📺]: Real-time multimodal AI travel guide with 3D map transitions
- IITM Paradox** [📺]: Steering Committee, Secretary & Core; built the org from scratch for a 22,000-student community.
- YC Startup School 2026** [📺]: Selected as one of 6,000 technical builders invited to YC's invite-only AI summit in SF.